ETHICAL HACKING PRACTICAL JOURNAL

Practical Write-ups

Name: _____

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Practical 1: WHOIS Lookup of a Website

Aim

To perform a WHOIS lookup and collect domain registration information of a website using DomainTools.

Theory

WHOIS is a protocol used to find information about a registered domain name. It can show the registrar, creation date, expiry date, name servers and related domain details.

Procedure

9. Open Google Chrome.
10. Visit https://whois.domaintools.com/.
11. Enter the domain name facebook.com in the search box.
12. Click the Search button.
13. Observe the WHOIS record displayed on the screen.
14. Note the registrar, creation date, expiry date, IP address and name servers.
15. Take a screenshot of the results.

Result

Successfully performed a WHOIS lookup of facebook.com using DomainTools and viewed its domain registration information.

Practical 2: RC4 Encryption and Decryption Using CrypTool

Aim

To perform encryption and decryption using the RC4 algorithm in CrypTool.

Theory

RC4 is a symmetric stream cipher. It uses the same secret key for encryption and decryption. Encryption converts plaintext into ciphertext, and decryption converts it back into the original message.

Procedure

16. Open CrypTool 1.4.42.
17. Create a new text document.
18. Enter sample plaintext, such as your name, class and subject.
19. Select the text.
20. Choose the RC4 encryption option from the Encrypt/Decrypt menu.
21. Enter a secret key and perform encryption.
22. Observe the encrypted ciphertext.
23. Select the ciphertext and choose RC4 decryption.
24. Enter the same secret key.
25. Verify that the original plaintext is recovered.

Note

Use the same key for encryption and decryption. RC4 is an outdated cipher and should be used only for this educational demonstration.

Result

Successfully performed RC4 encryption and decryption in CrypTool. The original message was recovered using the same key.

Practical 3: Installation and Configuration of Kali Linux Using VirtualBox

Aim

To import and configure Kali Linux in Oracle VirtualBox.

Theory

Kali Linux is a Debian-based operating system used for cybersecurity learning and penetration testing. Oracle VirtualBox allows Kali Linux to run as a virtual machine on a computer.

Procedure

26. Install Oracle VirtualBox on the computer.
27. Download the official Kali Linux VirtualBox image.
28. Open Oracle VirtualBox Manager.
29. Click Open and select the downloaded Kali virtual machine file.
30. Select the imported Kali Linux virtual machine.
31. Check its memory, processor and storage settings.
32. Check that the network adapter is configured as required; NAT is suitable for basic internet access.
33. Click Start to launch Kali Linux.
34. Complete the login and observe the Kali Linux desktop.

Result

Successfully imported and configured Kali Linux in Oracle VirtualBox.

Practical 4: Network Scanning Using Nmap

Aim

To scan a computer or authorised test system using Nmap and identify open ports and services.

Theory

Nmap (Network Mapper) is a network scanning tool used to discover hosts, open ports and running services. It helps administrators understand the network services available on a system.

Procedure

35. Start Kali Linux and open the Terminal.
36. Ensure that Nmap is installed.
37. Use an authorised target, such as your own computer or a lab machine.
38. Enter the command nmap 127.0.0.1 to scan the local computer.
39. Observe the list of open ports and their states.
40. For service information, use nmap -sV 127.0.0.1.
41. Note the port numbers, states and detected services.
42. Take a screenshot of the scan results.

Note

Scan only systems that you own or have permission to test. The results depend on which services are running.

Result

Successfully performed a basic Nmap scan on an authorised local target and observed the available ports and services.

Practical 5: Capture and Analysis of Network Traffic Using Wireshark

Aim

To capture and analyse network traffic using Wireshark.

Theory

Wireshark is a network packet analyser used to capture and inspect network packets. It can display protocol details, source and destination addresses, requests and responses.

Procedure

43. Open Wireshark.
44. Select the Wi-Fi or other active network interface.
45. Start packet capture.
46. Open an authorised HTTP test page or local lab login page in a browser.
47. Enter dummy test credentials and submit the form.
48. Return to Wireshark and stop the capture.
49. Enter http in the display filter, if the test traffic uses HTTP.
50. Locate the relevant request packet.
51. Right-click the packet and select Follow, then TCP Stream.
52. Observe the request and server response, then take a screenshot.

Note

Use only an authorised test page and dummy credentials. HTTP is unencrypted and may expose submitted form data. HTTPS traffic is encrypted.

Result

Successfully captured and analysed network traffic using Wireshark and observed request and response details.

Practical 6: Installation and Configuration of DVWA Using XAMPP (DVWA A)

Aim

To install, configure and run Damn Vulnerable Web Application (DVWA) using XAMPP.

Theory

DVWA is an intentionally vulnerable web application used to learn web security in a controlled environment. It includes practice modules for vulnerabilities such as SQL Injection and Cross-Site Scripting (XSS).

Procedure

53. Install and open XAMPP Control Panel.
54. Start the Apache and MySQL services.
55. Extract the DVWA folder into C:\xampp\htdocs.
56. Open a browser and visit http://localhost/DVWA/.
57. Log in using the configured DVWA administrator account. The default credentials are admin and password if they have not been changed.
58. Open Setup / Reset DB.
59. Click Create / Reset Database.
60. Open DVWA Security, select Low for the lab exercise and submit the setting.
61. Open a vulnerability module and verify that it loads.
62. Take screenshots of the setup and running application.

Note

Keep DVWA on localhost or an isolated lab machine. Never expose this intentionally vulnerable application to the public internet.

Result

Successfully installed and configured DVWA using XAMPP in a local testing environment.

Practical 7: SQL Injection Testing Using DVWA (DVWA B)

Aim

To demonstrate SQL Injection on the authorised DVWA practice application.

Theory

SQL Injection is a web application vulnerability in which unsafe user input changes the structure of a database query. It may allow unauthorised access to database records.

Procedure

63. Start Apache and MySQL in XAMPP.
64. Open http://localhost/DVWA/ and log in.
65. Set the DVWA security level to Low for the controlled lab exercise.
66. Select SQL Injection from the left-side menu.
67. Enter the normal test ID 1 and click Submit to view the basic result.
68. In the local DVWA lab only, enter the demonstration input 1' or true;#.
69. Click Submit and observe the records displayed.
70. Take a screenshot of the result.

Note

Perform this exercise only on your own local DVWA installation or another explicitly authorised training environment.

Result

Successfully demonstrated SQL Injection in the DVWA local lab and observed multiple database records.

Practical 8: Creation of a Simple Text Logger Using Python and Tkinter

Aim

To create a simple GUI application that saves text entered by the user into a file with the date and time.

Theory

Tkinter is Python's GUI library. It can create windows, text boxes and buttons. Python file handling can save user-entered text in a text file. This practical is a manual text-entry logger, not a system-wide keyboard monitor.

Procedure

71. Open Python IDLE or VS Code.
72. Import the Tkinter library and datetime module.
73. Create a GUI window using Tkinter.
74. Add a text box for entering sample text.
75. Create a Save button.
76. Write a save function that appends the entered text and current date and time to key_log.txt.
77. Run the program.
78. Enter sample text in the text box.
79. Click Save.
80. Open key_log.txt and verify the saved text and timestamp.

Note

This practical records only text deliberately entered into the application's text box. Do not use hidden keystroke capture or record other people's input.

Result

Successfully created a simple text-logging GUI using Python and Tkinter. The entered text was saved in key_log.txt with date and time.